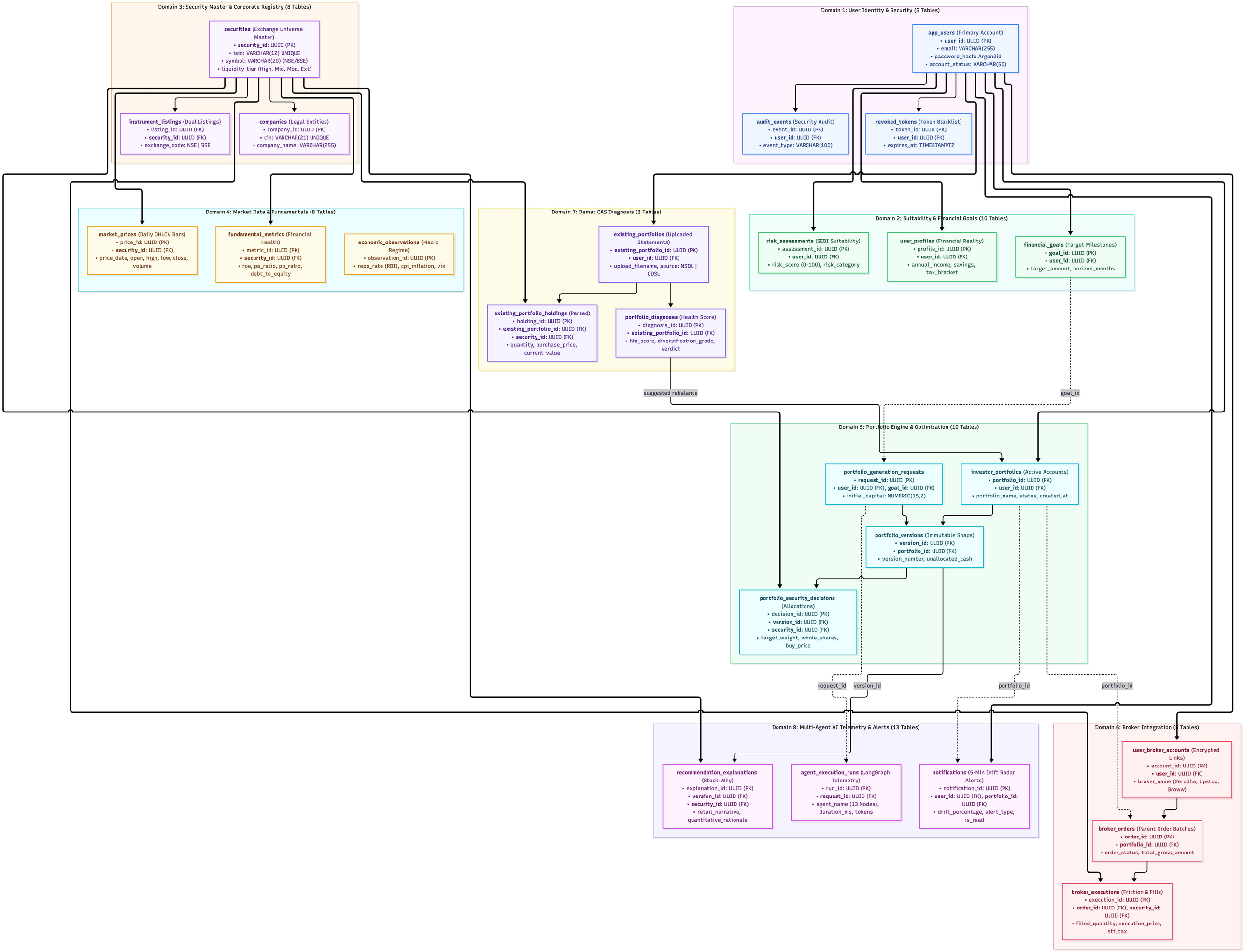

Domain 3: Security Master & Corporate Registry (8 Tables)
securities (Exchange Universe Master)
▪ security_id: UUID (PK)
▪ isin: VARCHAR(12) UNIQUE
▪ symbol: VARCHAR(20) (NSE/BSE)
▪ liquidity_tier (High, Mid, Mod, Ext)
instrument_listings (Dual Listings)
▪ listing_id: UUID (PK)
▪ security_id: UUID (FK)
▪ exchange_code: NSE | BSE
companies (Legal Entities)
▪ company_id: UUID (PK)
▪ cin: VARCHAR(21) UNIQUE
▪ company_name: VARCHAR(255)
Domain 1: User Identity & Security (5 Tables)
app_users (Primary Account)
▪ user_id: UUID (PK)
▪ email: VARCHAR(255)
▪ password_hash: Argon2id
▪ account_status: VARCHAR(50)
audit_events (Security Audit)
▪ event_id: UUID (PK)
▪ user_id: UUID (FK)
▪ event_type: VARCHAR(100)
revoked_tokens (Token Blacklist)
▪ token_id: UUID (PK)
▪ user_id: UUID (FK)
▪ expires_at: TIMESTAMPTZ
Domain 4: Market Data & Fundamentals (8 Tables)
market_prices (Daily OHLCV Bars)
▪ price_id: UUID (PK)
▪ security_id: UUID (FK)
▪ price_date, open, high, low, close, volume
fundamental_metrics (Financial Health)
▪ metric_id: UUID (PK)
▪ security_id: UUID (FK)
▪ roe, pe_ratio, pb_ratio, debt_to_equity
economic_observations (Macro Regime)
▪ observation_id: UUID (PK)
▪ repo_rate (RBI), cpi_inflation, vix
Domain 7: Demat CAS Diagnosis (3 Tables)
existing_portfolios (Uploaded Statements)
▪ existing_portfolio_id: UUID (PK)
▪ user_id: UUID (FK)
▪ upload_filename, source: NSDL | CDSL
existing_portfolio_holdings (Parsed)
▪ holding_id: UUID (PK)
▪ existing_portfolio_id: UUID (FK)
▪ security_id: UUID (FK)
▪ quantity, purchase_price, current_value
portfolio_diagnoses (Health Score)
▪ diagnosis_id: UUID (PK)
▪ existing_portfolio_id: UUID (FK)
▪ hhi_score, diversification_grade, verdict
Domain 2: Suitability & Financial Goals (10 Tables)
risk_assessments (SEBI Suitability)
▪ assessment_id: UUID (PK)
▪ user_id: UUID (FK)
▪ risk_score (0-100), risk_category
user_profiles (Financial Reality)
▪ profile_id: UUID (PK)
▪ user_id: UUID (FK)
▪ annual_income, savings, tax_bracket
financial_goals (Target Milestones)
▪ goal_id: UUID (PK)
▪ user_id: UUID (FK)
▪ target_amount, horizon_months
suggested rebalance
goal_id
Domain 5: Portfolio Engine & Optimization (10 Tables)
portfolio_generation_requests
▪ request_id: UUID (PK)
▪ user_id: UUID (FK), goal_id: UUID (FK)
▪ initial_capital: NUMERIC(15,2)
investor_portfolios (Active Accounts)
▪ portfolio_id: UUID (PK)
▪ user_id: UUID (FK)
▪ portfolio_name, status, created_at
portfolio_versions (Immutable Snaps)
▪ version_id: UUID (PK)
▪ portfolio_id: UUID (FK)
▪ version_number, unallocated_cash
portfolio_security_decisions (Allocations)
▪ decision_id: UUID (PK)
▪ version_id: UUID (FK)
▪ security_id: UUID (FK)
▪ target_weight, whole_shares, buy_price
request_id
version_id
portfolio_id
portfolio_id
Domain 8: Multi-Agent AI Telemetry & Alerts (13 Tables)
recommendation_explanations (Stock-Why)
▪ explanation_id: UUID (PK)
▪ version_id: UUID (FK)
▪ security_id: UUID (FK)
▪ retail_narrative, quantitative_rationale
agent_execution_runs (LangGraph Telemetry)
▪ run_id: UUID (PK)
▪ request_id: UUID (FK)
▪ agent_name (13 Nodes), duration_ms, tokens
notifications (5-Min Drift Radar Alerts)
▪ notification_id: UUID (PK)
▪ user_id: UUID (FK), portfolio_id: UUID (FK)
▪ drift_percentage, alert_type, is_read
Domain 6: Broker Integration (3 Tables)
user_broker_accounts (Encrypted Links)
▪ account_id: UUID (PK)
▪ user_id: UUID (FK)
▪ broker_name (Zerodha, Upstox, Groww)
broker_orders (Parent Order Batches)
▪ order_id: UUID (PK)
▪ portfolio_id: UUID (FK)
▪ order_status, total_gross_amount
broker_executions (Friction & Fills)
▪ execution_id: UUID (PK)
▪ order_id: UUID (FK), security_id: UUID (FK)
▪ filled_quantity, execution_price, stt_tax